

StudySync Project Proposal

AI Study Buddy: An Interactive Study Companion for Students

Vision Statement

My goal is to design and develop an AI-assisted study application that turns students' course materials, such as notes and instructor-provided practice materials, into personalized study guides and practice questions. The application will also help students work through problems interactively by explaining concepts and identifying gaps in their understanding rather than simply providing answers. Through this project, I will strengthen my software development skills while gaining practical experience integrating AI into a user-focused application, including prompt and context design, AI output evaluation, and responsible AI use in education.

Motivation

My main motivation for this project comes from my own experience as an online student. While I have had some very involved professors, I have also taken asynchronous courses where there is little interaction with the professor or other students. Learning difficult material independently can feel isolating, especially when entering a new field, and it can be difficult to know what to study, whether I truly understand a concept, or where the gaps in my knowledge are.

Throughout my degree, using AI as a "study buddy" has been extremely valuable. It has helped me organize my notes, review concepts in different ways, generate additional practice, and better understand material when I am struggling with it. I want to explore how that kind of support could be built into a purposeful study tool that supplements a student's coursework rather than completing it for them. At the same time, developing the application will strengthen my programming and AI development skills and give me a portfolio project centered on a problem I genuinely care about solving.

Risks to Project Completion

- **Scope:** StudySync could easily grow beyond what can reasonably be completed within the 45-hour project limit. Features beyond the core MVP, such as supporting multiple document types, generating full practice exams, user accounts, saved study materials, and additional personalization could make the application too large for the available time.
- **Unfamiliar Technology:** This project will require me to work with AI tools and technologies that I have not previously used in a complete application. The time required to learn, integrate, and troubleshoot these technologies could slow development or require changes to the original design.
- **Competing Responsibilities:** I am completing this project alongside my other university courses and professional work. Unexpected increases in coursework or work responsibilities could reduce the amount of time I am able to dedicate to StudySync during a particular week.

Mitigation Strategies

- **Scope:** I will define a clear minimum viable product (MVP) before beginning development and prioritize the features required for that version. Additional features will be treated as optional stretch goals and can be removed as the semester progresses without preventing completion of the core application.
- **Unfamiliar Technology:** I will time-box research and experimentation with unfamiliar technologies so that I do not spend too much of the project trying to solve a single technical problem. When I am stuck, I will seek help through available resources and request one-on-one meetings when necessary. If a technology proves too complex for the project's timeframe, I will adjust the implementation rather than allowing it to prevent progress.
- **Competing Responsibilities:** I will use project-management practices from my Software Design coursework, including a Jira board and weekly development milestones, to break the project into manageable tasks and monitor my progress. If coursework or professional responsibilities reduce my available time, I will reprioritize the remaining tasks and cut optional features before compromising the MVP.

Project Assessments

The project will be considered successful if I complete a functional minimum viable product of StudySync within the approximately 45 hours available for the project. The MVP will be evaluated using the following criteria:

- **Course Material Input:** A user can provide course notes or other study materials that StudySync can use as context.
- **Study Material Generation:** StudySync can use the provided course material to generate a study guide or set of practice questions that is based on the supplied material.
- **Interactive Study Support:** A user can interact with StudySync to ask questions or request explanations based on the provided course material.
- **AI Output Evaluation:** I will test the application's generated study materials and responses using sample course materials and document the results, including examples of outputs that work well and areas that need improvement.
- **Project Documentation:** I will document the development process, major design decisions, challenges, and final results on my project portfolio website.
- **Professional Growth:** At the end of the project, I will reflect on the software development and AI integration skills I developed and identify areas that I would continue improving if StudySync were developed further.

Project Portfolio

StudySync will be documented throughout the semester on my existing professional portfolio website:

Portfolio Website: kflick3r.github.io

StudySync GitHub Repository: github.com/kflick3r/StudySync